

Weitong Zhang

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PROFESSIONAL EXPERIENCE

University of North Carolina, Chapel Hill

Tenure-Track Assistant Professor, School of Data Science and Society

Chapel Hill, NC, USA

2024–Current

EDUCATION

University of California, Los Angeles

Ph.D. in Computer Science, Advisor: Quanquan Gu, Dissertation Year Award

Los Angeles, CA, USA

2019–2024

Tsinghua University

B.E. in Automation, Outstanding Graduate Award

Beijing, China

2015–2019

RESEARCH INTEREST

My research focuses on machine learning, with an emphasis on **reinforcement learning** (RL). I aim to advance the theoretical foundations of RL and broaden its applicability including large language models, agentic systems, **embodied AI** and generative modeling. I am also working on developing **AI for scientific discovery**, particularly in the **self-driving laboratories** in chemistry using specific-designed RL algorithms and modern machine learning models.

SCHOLARSHIPS AND AWARDS

- UNC Junior Faculty Award (\$10,000) 2026
- AAAI New Faculty Award 2025
- UCLA Dissertation Year Fellowship 2023
- UCLA Summer Mentored Research Fellowship 2021
- Qualcomm Scholarship 2017–2018
- Finalist in Mathematical Contests in Modeling (MCM) 2017
- China National Scholarship 2016
- Tsinghua Outstanding Freshmen Scholarship 2015
- Golden Prize for Chinese Physics Olympiad (CPHO) 2014

RESEARCH GRANTS AND EXTERNAL SUPPORT

- NVIDIA Academic Program 2026, Federated Agentic Alignment: Hybrid Transfer RL with Local Self-Play. (Single PI)
- UNC – Tübingen Joint Grant: Flow-Based Generative Models for Data-Efficient Imitation Learning (Lead PI, share \$10,000)
- NVIDIA Academic Program 2-25, Scalable Energy Guided Graph Diffusion Model for Accelerated Drug Design (Lead PI)
- Google Cloud Research Credit, Exact Energy Guided Generative Modeling for De-novo Drug Design

TEACHING

- **Instructor** at University of North Carolina at Chapel Hill Fall 2026
DATA 703: Introduction to Programming ([Course Website](#))
- **Instructor** at University of North Carolina at Chapel Hill Spring 2026
DATA 522: Practical Deep Learning Systems ([Course Website](#))
- **Instructor** at University of North Carolina at Chapel Hill Spring 2025
DATA 890: Special Topics: Decision Making and Reinforcement Learning ([Course Website](#))
- **Teaching Assistant** at University of California, Los Angeles Spring 2021, Fall 2021
Introductory Digital Design Laboratory (CS M152A) ([Course Website 21 Fall](#), [Course Website 21 Spring](#))
- **Teaching Assistant** at University of California, Los Angeles Winter 2022, Winter 2023
Fundamentals of Artificial Intelligence (CS 161) ([Course Website 22 Winter](#), [Course Website 23 Winter](#))

PROFESSIONAL SERVICES

Conference Area Chair/Senior Program Committee

- [International Conference on Machine Learning \(ICML\)](#), 2025, 2026
- [International Conference on Learning Representations \(ICLR\)](#), 2026
- [Neural Information Processing Systems \(NeurIPS\)](#), 2026
- [Transactions on Machine Learning Research \(TMLR\)](#)

Conference Reviewer / Program Committee

- [International Conference on Learning Representations \(ICLR\)](#), 2020, 2021, 2022, 2023, 2024, 2025
- [Neural Information Processing Systems \(NeurIPS\)](#), 2021, 2022, 2023, 2024
- [International Conference on Machine Learning \(ICML\)](#), 2021, 2022, 2023, 2024
- [International Joint Conference on Artificial Intelligence \(IJCAI\)](#), 2020, 2021, 2022, 2023, 2024
- [International Conference on Artificial Intelligence and Statistics \(AISTATS\)](#), 2022, 2023, 2024

Journal Reviewer

- [Journal of Artificial Intelligence Research \(JAIR\)](#)
- [Transactions on Machine Learning Research \(TMLR\)](#)
- [PLOS ONE](#)
- [PLOS Global Public Health](#)

INVITED TALKS & PANEL DISCUSSIONS

- Exact Energy Guidance in Diffusion Models and offline RL
 - Invited Seminar, BCBB Seminar Series, NIH/NIEHS 2025-11-04
 - INFORMS 2025, Emerging Technology in Integrated Risk Management 2025-10-28
 - Research Seminar, UNC School of Dentistry, Host: Dr. Di Wu 2025-04-13
 - Research in Progress, UNC School of Medicine 2025-03-15
 - Samsung AI seminar 2025-02-28

- Department of Computer Science, Indiana University Bloomington 2024-12-05
- Research Seminar, NVIDIA-Astrazeneca 2024-11-08
- Duke Industry Statistics Symposium 2025-04-10
Panelist: S3D – Challenges and Opportunities of ML/AL in Drug Development
- Introduction to Reinforcement Learning: from Bayesian Optimization to Bandits 2024-08-29
Research Seminar, UCLA Department of Chemistry, Host: Dr. Chong Liu
- Uncertainty-Aware Unsupervised and Robust Reinforcement Learning 2024-08-23
Research Seminar, Department of Biostatistics & Bioinformatics, Duke University, Host: Dr. Pan Xu

PUBLICATIONS

* indicates equal contribution

- [Hua+26] Aijian Huang, Jared S. Stanley, Yi-An Lai, Hongyuan Sheng, Haiyuan Zou, Hao Ming Chen, Cyrille Costentin, Long Qi, Jenny Y. Yang, **Weitong Zhang**, and Chong Liu. “Latent mechanism mapping in self-driving molecular electrocatalysis”. In: *Submitted preprint* (2026).
- [Lai+26] Yi-An Lai, Muxin Xiong, Jingwen Sun, Aijian Huang, Max Van Fleet, Hongyuan Sheng, Jennifer Kim, F. Kuo, Tai Ying Lai, Xuian-Rou Lin, Yu-An Chen, **Weitong Zhang**, Hao Ming Chen, and Chong Liu. “Chemistry-infused bandit framework for interpretable autonomous campaign of electrocatalysis”. In: *Submitted preprint* (2026).
- [Li+26] Shangzhe Li, Xuchao Zhang, Chetan Bansal, and **Weitong Zhang**. “Your Self-Play Algorithm is Secretly an Adversarial Imitator: Understanding LLM Self-Play through the Lens of Imitation Learning”. In: *arXiv preprint arXiv:2602.01357* (2026). arXiv: [2602.01357](#) [[cs.LG](#)].
- [LZZ26] Shangzhe Li, Dongruo Zhou, and **Weitong Zhang**. “Near-Optimal Second-Order Guarantees for Model-Based Adversarial Imitation Learning”. In: *International Conference on Learning Representations*. 2026.
- [LW26] Shangzhe Liang and **Weitong Zhang**. “Provably Efficient Offline-to-Online Value Adaptation with General Function Approximation”. In: *arXiv preprint arXiv:2604.13966* (2026). arXiv: [2604.13966](#) [[cs.LG](#)].
- [LLW26] Yiyang Liang, Sifei Liu, and **Weitong Zhang**. “Moment Matching Q-Learning”. In: *International Conference on Machine Learning (ICML 2026)*. Accepted by ICML 2026. 2026. arXiv: [2605.29033](#) [[cs.LG](#)].
- [QLZ26] Yongtao Qu, Shangzhe Li, and **Weitong Zhang**. “Imitation from Observations with Trajectory-Level Generative Embeddings”. In: *arXiv preprint arXiv:2601.00452* (2026).
- [Sun+26] Jingwen Sun, **Weitong Zhang**, Qingyue Zhao, Quanquan Gu, and Chong Liu. “EIS-GPT: Transformer-based generation of equivalent circuit for electrochemical impedance spectroscopy”. In: *ChemRxiv preprint* (2026).
- [Wei26] **Weitong Zhang**. “Reinforcement Learning Without Explicit Rewards: Theory and Practice”. In: *Proceedings of the AAAI Conference on Artificial Intelligence* 40.47 (2026), pp. 39847–39847.

- [YWW26] Yuxiao Yang, Xiaoyun Wang, and **Weitong Zhang**. “OGLS-SD: On-Policy Self-Distillation with Outcome-Guided Logit Steering for LLM Reasoning”. In: *arXiv preprint arXiv:2605.12400* (2026). arXiv: [2605.12400 \[cs.LG\]](#).
- [YW26] Yuxiao Yang and **Weitong Zhang**. “Return-to-Go Is More Than a Number: Q-Guided Alignment for Return-Conditioned Supervised Learning”. In: *International Conference on Machine Learning (ICML 2026)*. Accepted by ICML 2026. 2026. arXiv: [2605.29028 \[cs.LG\]](#).
- [Yu+26a] Tianrun Yu, Yuxiao Yang, Zhaoyang Wang, Kaixiang Zhao, Porter Jenkins, Xuchao Zhang, Chetan Bansal, Huaxiu Yao, and **Weitong Zhang**. “Provable and Practical In-Context Policy Optimization for Self-Improvement”. In: *International Conference on Learning Representations (ICLR 2026)*. Accepted by ICLR 2026. 2026. arXiv: [2603.01335 \[cs.LG\]](#).
- [Yu+26b] Tianrun Yu, Kaixiang Zhao, Chih-Chun Chen, Amanda Hughes, Taylor W Killian, Fenglong Ma, **Weitong Zhang**, and Porter Jenkins. “LARK: Learnability-Grounded Trajectory Selection for Efficient Reasoning Distillation”. In: *arXiv preprint arXiv:2605.30651* (2026). arXiv: [2605.30651 \[cs.LG\]](#).
- [Lan+25] Yifan Lan, Yuanpu Cao, **Weitong Zhang**, Lu Lin, and Jinghui Chen. “Phi: Preference Hijacking in Multi-modal Large Language Models at Inference Time”. In: *The 2025 Conference on Empirical Methods in Natural Language Processing (oral)*. 2025.
- [Sun+25] Jingwen Sun, **Weitong Zhang**, Yuanzhou Chen, Benjamin B Hoar, Hongyuan Sheng, Jenny Y Yang, Quanquan Gu, and Chong Liu. “Inquiry into the Appropriate Data Preprocessing of Electrochemical Impedance Spectroscopy for Machine Learning”. In: *The Journal of Physical Chemistry C* (2025).
- [Wan+25a] Yilin Wang, Shangzhe Li, Haoyi Niu, Zhiao Huang, **Weitong Zhang**, and Hao Su. “A Recipe for Efficient Sim-to-Real Transfer in Manipulation with Online Imitation-Pretrained World Models”. In: *arXiv preprint arXiv:2510.02538* (2025).
- [Wan+25b] Zhaoyang Wang, Weilei He, Zhiyuan Liang, Xuchao Zhang, Chetan Bansal, Ying Wei, **Weitong Zhang**, and Huaxiu Yao. “CREAM: Consistency Regularized Self-Rewarding Language Models”. In: *International Conference on Learning Representations*. 2025.
- [ZZG25] Shiyuan Zhang, **Weitong Zhang**, and Quanquan Gu. “Energy-Weighted Flow Matching for Offline Reinforcement Learning”. In: *International Conference on Learning Representations*. 2025.
- [Zha+25] Linxi Zhao, Yihe Deng, **Weitong Zhang**, and Quanquan Gu. “Mitigating Object Hallucination in Large Vision-Language Models via Image-Grounded Guidance”. In: *Forty-second International Conference on Machine Learning (spotlight)*. 2025.
- [Zho+25] Yiyang Zhou, Zhaoyang Wang, Tianle Wang, Shangyu Xing, Peng Xia, Bo Li, Kaiyuan Zheng, Zijian Zhang, Zhaorun Chen, Wenhao Zheng, Xuchao Zhang, Chetan Bansal, **Weitong Zhang**, Ying Wei, Mohit Bansal, and Huaxiu Yao. “AnyPrefer: An Automatic Framework for Preference Data Synthesis”. In: *International Conference on Learning Representations*. 2025.

- [BZG24] Alexander Braverman, **Weitong Zhang**, and Quanquan Gu. “Mitigating Hallucination in Large Language Models with Explanatory Prompting”. In: *Statistical Foundations of LLMs and Foundation Models (NeurIPS 2024 Workshop)*. 2024.
- [Hoa+24] Benjamin B Hoar, **Weitong Zhang**, Yuanzhou Chen, Jingwen Sun, Hongyuan Sheng, Yucheng Zhang, Yisi Chen, Jenny Y Yang, Cyrille Costentin, Quanquan Gu, et al. “Redox-Detecting Deep Learning for Mechanism Discernment in Cyclic Voltammograms of Multiple Redox Events”. In: *ACS electrochemistry* 1.1 (2024), pp. 52–62.
- [Hua+24] Zijie Huang, Jeehyun Hwang, Junkai Zhang, Jinwoo Baik, **Weitong Zhang**, Dominik Wodarz, Yizhou Sun, Quanquan Gu, and Wei Wang. “Causal Graph ODE: Continuous Treatment Effect Modeling in Multi-agent Dynamical Systems”. In: *Proceedings of the ACM on Web Conference 2024*. 2024, pp. 4607–4617.
- [She+24] Hongyuan Sheng, Jingwen Sun, Oliver Rodríguez, Benjamin B Hoar, **Weitong Zhang**, Danlei Xiang, Tianhua Tang, Avijit Hazra, Daniel S Min, Abigail G Doyle, et al. “Autonomous closed-loop mechanistic investigation of molecular electrochemistry via automation”. In: *Nature Communications* 15.1 (2024), p. 2781.
- [Zha+24a] Junkai Zhang, **Weitong Zhang**, Dongruo Zhou, and Quanquan Gu. “Uncertainty-Aware Reward-Free Exploration with General Function Approximation”. In: *Forty-first International Conference on Machine Learning*. 2024.
- [Zha+24b] **Weitong Zhang**, Zhiyuan Fan, Jiafan He, and Quanquan Gu. “Achieving Constant Regret in Linear Markov Decision Processes”. In: *The Thirty-eighth Annual Conference on Neural Information Processing Systems*. 2024.
- [Zhe+24] Wenhao Zheng, Yixiao Chen, **Weitong Zhang**, Souvik Kundu, Yun Li, Zhengzhong Liu, Eric P Xing, Hongyi Wang, and Huaxiu Yao. “CITER: Collaborative Inference for Efficient Large Language Model Decoding with Token-Level Routing”. In: *Adaptive Foundation Models: Evolving AI for Personalized and Efficient Learning*. 2024.
- [Den+23] Yihe Deng, **Weitong Zhang**, Zixiang Chen, and Quanquan Gu. “Rephrase and respond: Let large language models ask better questions for themselves”. In: *arXiv preprint arXiv:2311.04205* (2023).
- [Ji+23] Kaixuan Ji, Qingyue Zhao, Jiafan He, **Weitong Zhang**, and Quanquan Gu. “Horizon-free Reinforcement Learning in Adversarial Linear Mixture MDPs”. In: *The Twelfth International Conference on Learning Representations*. 2023.
- [ZZG23] Junkai Zhang, **Weitong Zhang**, and Quanquan Gu. “Optimal horizon-free reward-free exploration for linear mixture mdps”. In: *International Conference on Machine Learning*. PMLR. 2023, pp. 41902–41930.
- [Zha+23a] **Weitong Zhang**, Jiafan He, Zhiyuan Fan, and Quanquan Gu. “On the interplay between misspecification and sub-optimality gap in linear contextual bandits”. In: *International Conference on Machine Learning*. PMLR. 2023, pp. 41111–41132.
- [Zha+23b] **Weitong Zhang**, Jiafan He, Dongruo Zhou, Q Gu, and A Zhang. “Provably efficient representation selection in low-rank Markov decision processes: from online to offline RL”. In: *Uncertainty in Artificial Intelligence*. PMLR. 2023, pp. 2488–2497.

- [Zha+23c] **Weitong Zhang**, Xiaoyun Wang, Weili Nie, Joe Eaton, Brad Rees, and Quanquan Gu. “MoleculeGPT: Instruction Following Large Language Models for Molecular Property Prediction”. In: *NeurIPS 2023 Workshop on New Frontiers of AI for Drug Discovery and Development*. 2023.
- [Zha+23d] **Weitong Zhang**, Xiaoyun Wang, Justin Smith, Joe Eaton, Brad Rees, and Quanquan Gu. “Diffmol: 3d structured molecule generation with discrete denoising diffusion probabilistic models”. In: *ICML 2023 Workshop on Structured Probabilistic Inference & Generative Modeling*. 2023.
- [Hoa+22] Benjamin B Hoar, **Weitong Zhang**, Shuangning Xu, Rana Deeba, Cyrille Costentin, Quanquan Gu, and Chong Liu. “Electrochemical mechanistic analysis from cyclic voltammograms based on deep learning”. In: *ACS Measurement Science Au* 2.6 (2022), pp. 595–604.
- [Jia+21] Yiling Jia, **Weitong Zhang**, Dongruo Zhou, Quanquan Gu, and Hongning Wang. “Learning Neural Contextual Bandits through Perturbed Rewards”. In: *International Conference on Learning Representations*. 2021.
- [ZZG21] **Weitong Zhang**, Dongruo Zhou, and Quanquan Gu. “Reward-free model-based reinforcement learning with linear function approximation”. In: *Advances in Neural Information Processing Systems* 34 (2021), pp. 1582–1593.
- [Wu+20] Yue Frank Wu, **Weitong Zhang**, Pan Xu, and Quanquan Gu. “A finite-time analysis of two time-scale actor-critic methods”. In: *Advances in Neural Information Processing Systems* 33 (2020), pp. 17617–17628.
- [Zha+20] **Weitong Zhang**, Dongruo Zhou, Lihong Li, and Quanquan Gu. “Neural Thompson Sampling”. In: *International Conference on Learning Representations*. 2020.
- [Zou+20] Difan Zou, Lingxiao Wang, Pan Xu, Jinghui Chen, **Weitong Zhang**, and Quanquan Gu. “Epidemic model guided machine learning for COVID-19 forecasts in the United States”. In: *MedRxiv* (2020), pp. 2020–05.
- [Liu+18] Shuai Liu, **Weitong Zhang**, Xiaojun Wu, Shuo Feng, Xin Pei, and Danya Yao. “A simulation system and speed guidance algorithms for intersection traffic control using connected vehicle technology”. In: *Tsinghua Science and Technology* 24.2 (2018), pp. 160–170.

Publications as group authors

- [Lop+24] Velma K Lopez, Estee Y Cramer, Robert Pagano, John M Drake, Eamon B O’Dea, Madeline Adee, Turgay Ayer, Jagpreet Chhatwal, Ozden O Dalgic, Mary A Ladd, et al. “Challenges of COVID-19 Case Forecasting in the US, 2020–2021”. In: *PLoS computational biology* 20.5 (2024), e1011200.
- [She+23] Katriona Shea, Rebecca K Borchering, William JM Probert, Emily Howerton, Tiffany L Bogich, Shou-Li Li, Willem G van Panhuis, Cecile Viboud, Ricardo Aguás, Artur A Belov, et al. “Multiple models for outbreak decision support in the face of uncertainty”. In: *Proceedings of the National Academy of Sciences* 120.18 (2023), e2207537120.

- [Cra+22] Estee Y Cramer, Evan L Ray, Velma K Lopez, Johannes Bracher, Andrea Brennen, Alvaro J Castro Rivadeneira, Aaron Gerding, Tilmann Gneiting, Katie H House, Yuxin Huang, et al. “Evaluation of individual and ensemble probabilistic forecasts of COVID-19 mortality in the United States”. In: *Proceedings of the National Academy of Sciences* 119.15 (2022), e2113561119.
- [Bra+21] Johannes Bracher, Daniel Wolffram, Jannik Deuschel, Konstantin Görgen, Jakob L Ketterer, Alexander Ullrich, Sam Abbott, Maria Vittoria Barbarossa, Dimitris Bertsimas, Sangeeta Bhatia, et al. “A pre-registered short-term forecasting study of COVID-19 in Germany and Poland during the second wave”. In: *Nature communications* 12.1 (2021), p. 5173.
- [Ray+20] Evan L Ray, Nutchana Wattanachit, Jarad Niemi, Abdul Hannan Kanji, Katie House, Estee Y Cramer, Johannes Bracher, Andrew Zheng, Teresa K Yamana, Xinyue Xiong, et al. “Ensemble forecasts of coronavirus disease 2019 (COVID-19) in the US”. In: *MedRxiv* (2020), pp. 2020–08.